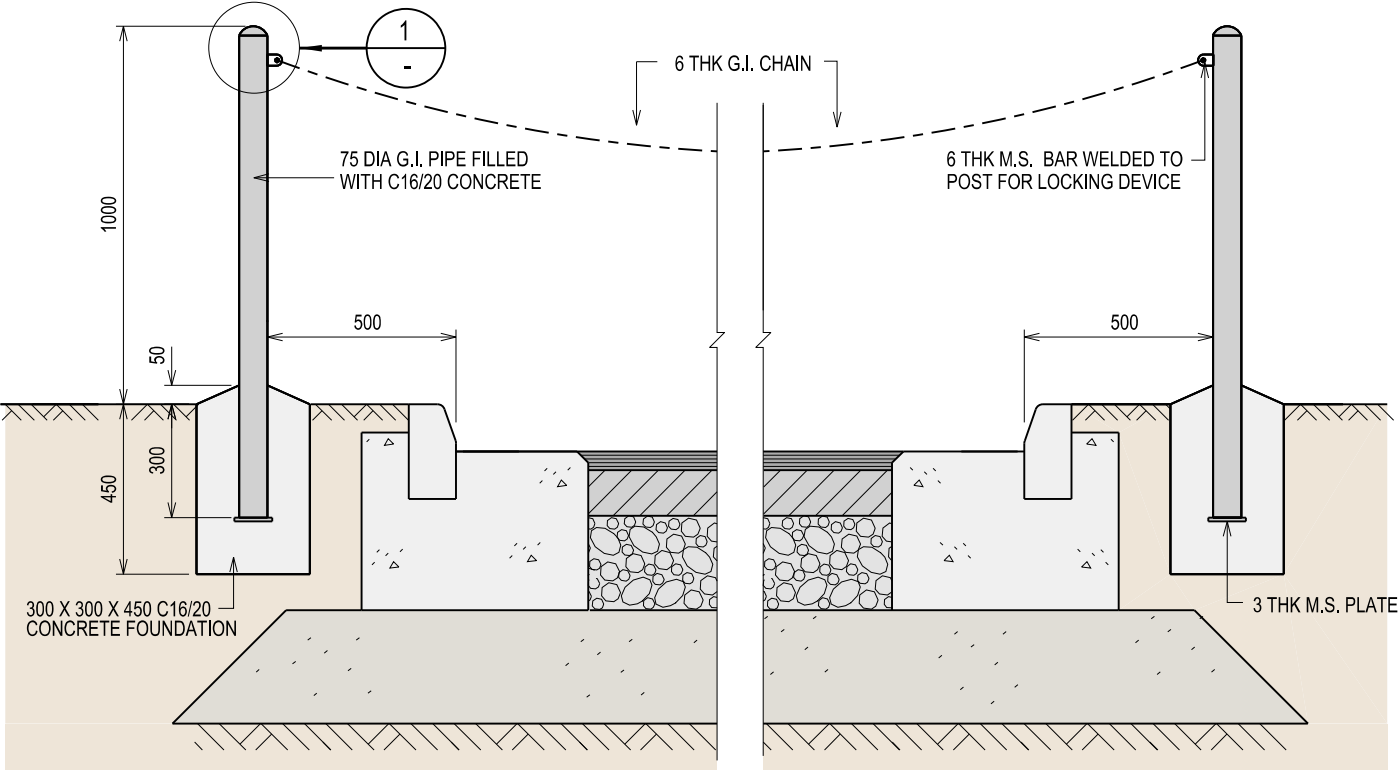


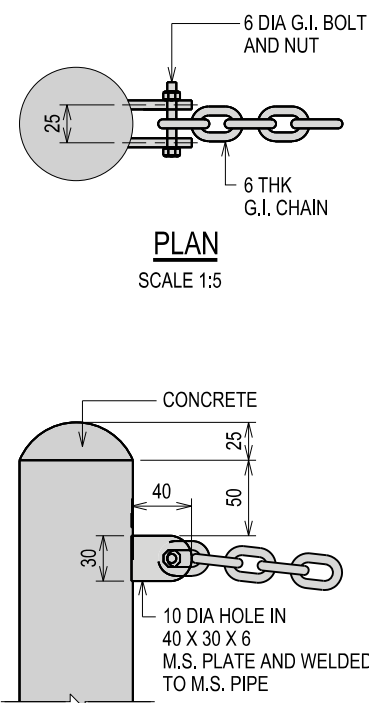
CHAPTER 14

BOLLARDS

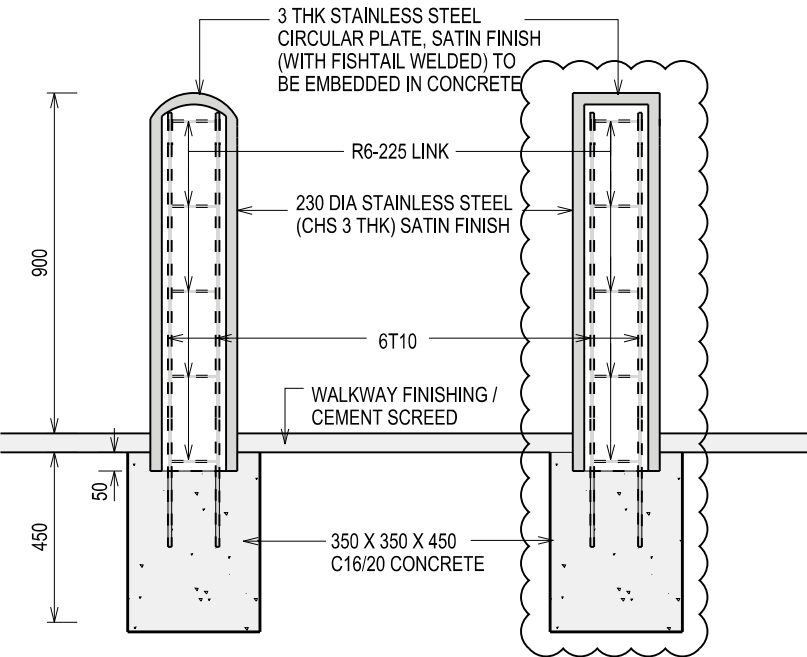
DRAWING TITLE	DRAWING NO.	REVISION
BOLLARDS	LTA/SDRE14/14/BOL1	B
SPRING LOADED POST & FLEXIBLE POST	LTA/SDRE14/14/BOL2	A
SAFETY BOLLARD AT BUS STOP - POSTED SPEED : 60km/h & 70km/h (SHEET 1 OF 2)	LTA/SDRE14/14/BOL3	A
SAFETY BOLLARD AT BUS STOP - POSTED SPEED : 60km/h & 70km/h (SHEET 2 OF 2)	LTA/SDRE14/14/BOL4	A
SAFETY BOLLARD AT TURFED AREA - POSTED SPEED : 60km/h & 70km/h	LTA/SDRE14/14/BOL5	A



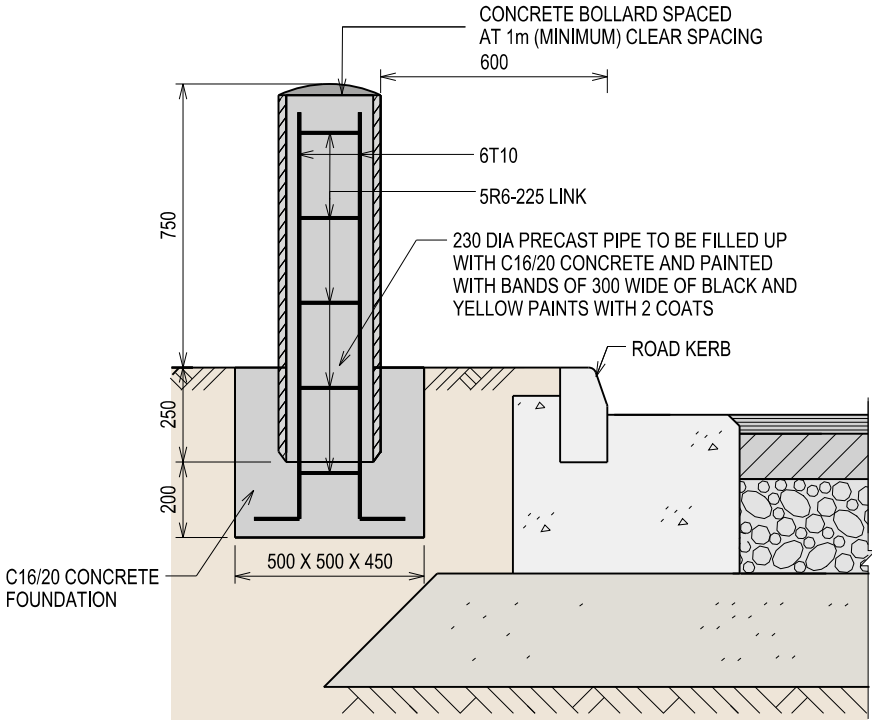
BOLLARD TO PREVENT VEHICLE ENTRY / PARKING
SCALE 1:20



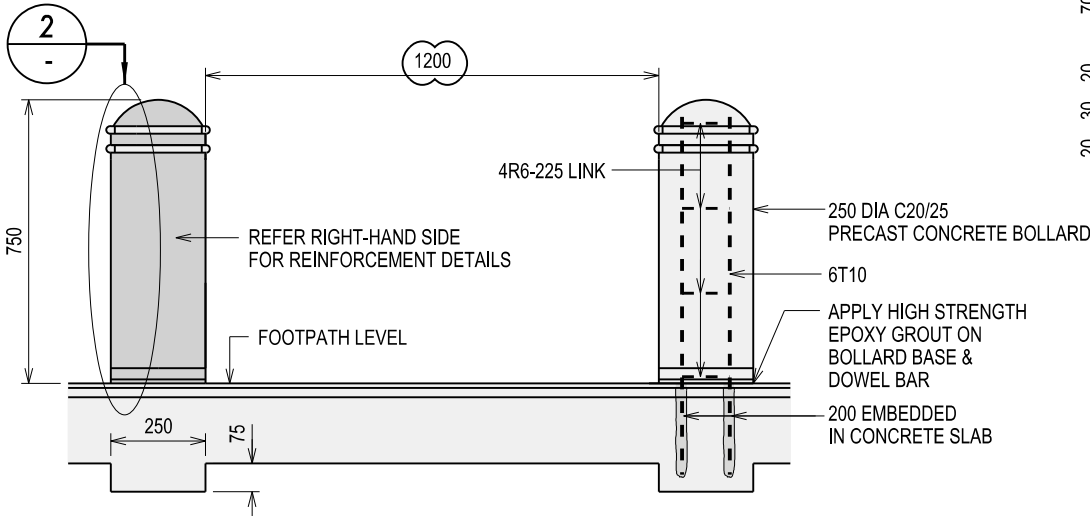
DETAIL 1
SCALE 1:5



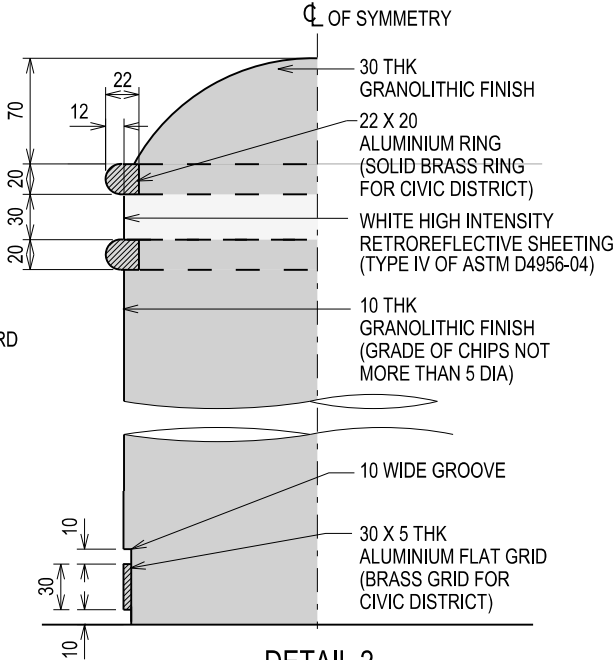
STAINLESS STEEL BOLLARD ON PAVEMENT
SCALE 1:20



CONCRETE BOLLARD ON GROUND
SCALE 1:20



CONCRETE BOLLARD ON PAVEMENT
(FOR CIVIC DISTRICT)
SCALE 1:20



DETAIL 2
SCALE 1:5

NOTES:
CONCRETE

1. Minimum concrete cover to nearest reinforcement shall be 50mm for the foundation and 30mm for others.

REINFORCEMENT

2. All steel reinforcement bars shall conform to BS 4449:1997 and SS 560.

STANDARD DETAIL

BOLLARDS

Land Transport Authority

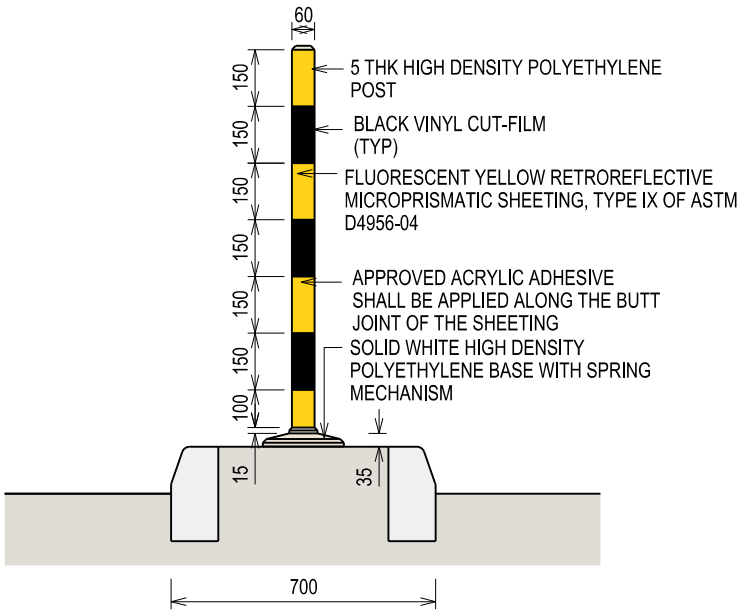
DRAWING NO. LTA/SDRE14/14/BOL1

REV. B

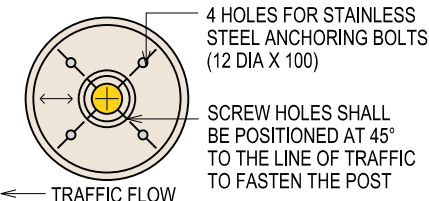
DATE OF ISSUE
1st APR 2014

SCALE
AS SHOWN

SHEET NO.
1 OF 1



SPRING LOADED POST
FOR NARROW DIVIDER (≤700mm WIDE)
SCALE 1:10



BASE - (PLAN VIEW)
SCALE 1:10

NOTE FOR INSTALLATION OF SPRING LOADED POST (SLP):

1. Spring loaded post shall comply to National Cooperative Highway Research Programme (NCHRP) Report 350, Category 1 with manufacturer's self-certification.
2. Spring loaded post shall consist of a spring mechanism which enables the post to rebound back to its original vertical position upon any impact and it is meant for installing on raised kerb / divider with a width not more than 700mm.
3. The base of the spring loaded post shall be secured to the concrete divider by minimum 4 nos. of stainless steel anchor bolts with a size of 12mm diameter x 100mm long.
4. One-year manufacturer's warranty shall be provided against permanent damage as a result of impact on the high density polyethylene (HDPE) post component.
5. Date of installation of the post shall be pasted on the post with the names of the supplier and the reflective sheeting.

GUIDELINES ON APPLICATION OF FLEXIBLE POST (FP):

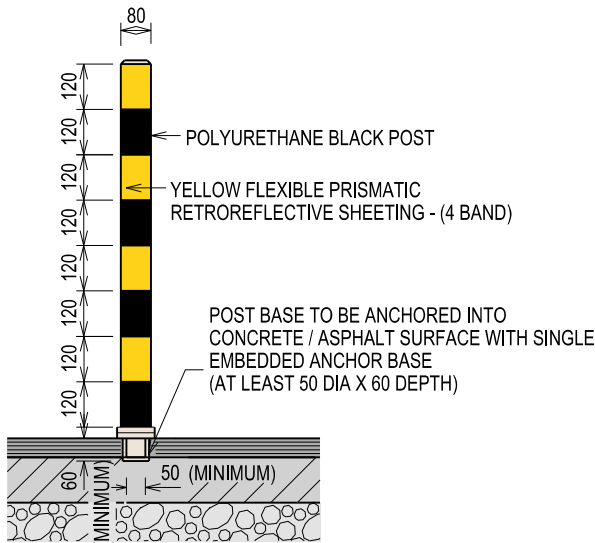
Flexible post can be applied on the following locations:

1. On road pavement:
 - a) Flexible post can be installed on road pavement if there are two or more damages on the existing spring loaded posts (slp) per year.
 - b) One-third length of the chevron marking area where the slp are installed, can be replaced with flexible post. for chevron marking areas on expressway and arterial road, flexible post can be spaced at 2.5m and 1.8m respectively and at the centre of the chevron marking.
 - c) They shall not be installed on top of the existing chevron marking.
2. At gore areas with crash cushion:
 - a) Flexible post can be installed in front of crash cushion if there are two or more damages on the crash cushion per year.
 - b) Installation of flexible post shall be confined to one-third length of the chevron marking starting from the crash cushion. spacing of flexible post shall be 2.5m and at the centre of the chevron marking.
 - c) They shall not be installed on top of the existing chevron marking.
3. On centre divider
 - a) If there are two or more damages on the slp per year on the centre divider, the damaged slp can be replaced with flexible post.

SPECIFICATIONS OF FLEXIBLE POST (FP)

1. The manufacturer is to certify that flexible post complies with the requirements of category 1, National Cooperative Highway Research Programme (NCHRP) Report 350 or equivalent. Category 1 includes small and lightweight items, such as channelizing and delineating devices that have been in common use for many years and are known to be tested to be crashworthy and years of demonstrable safe performance. these devices are allowed for use under the manufacturer's self certification.
2. Flexible post shall be made of polyurethane and capable of withstanding repeated impacts and wheel-overs in all directions. The post shall not deform more than 10 degrees from the vertical position and capable of retaining its structural performance when exposed to prevailing weather and traffic conditions within the three (3) years warranty period.
3. Flexible post shall be tested in an established laboratory with test facility such as flying wheel test or equivalent. the crash testing is to be recorded on videotape, and include a means, other than the test vehicle's speedometer, for determining the vehicle speed at the time of impact (range from 35 to 100 km/h) and to state the vehicle mass (at least an 800kg passenger car). In the manufacturer's certification, it is also required to state that the flexible post shall not pose any risk to impacting vehicle occupants.
4. Flexible post is to return to its original vertical orientation with quick rebound after every impact in any direction, therefore eliminating the need for re-straightening. It must be impact tested to at least 1,000 impacts without failure and bounce back to original vertical position.
5. Flexible post shall be in black and pasted with yellow flexible prismatic reflective sheeting which comply with ASTM E810 with minimum coefficient of retroreflection as follows:
6. The yellow flexible prismatic reflective sheeting shall comply with ASTM D4956-S2.2.2 flexibility under S2. reboundable sheeting requirements.
7. the epoxy resin mortar used to embed the post base into concrete / asphalt surface should have at least a compressive strength of 65n/mm², tensile strength of 13N/mm² and flexural strength of 35N/mmf.

OBSERVATION ANGLE	0.2°		0.5°	
ENTRANCE ANGLE	-4°	30°	-4°	30°
COEFFICIENT OF RETROFLECTION (CANDELAS PER LUX PER SQUARE METRE)	270	135	110	54



FLEXIBLE POST
SCALE 1:10

NOTES:
CONCRETE

1. Minimum concrete cover to nearest reinforcement shall be 50mm for the foundation and 30mm for others.

REINFORCEMENT

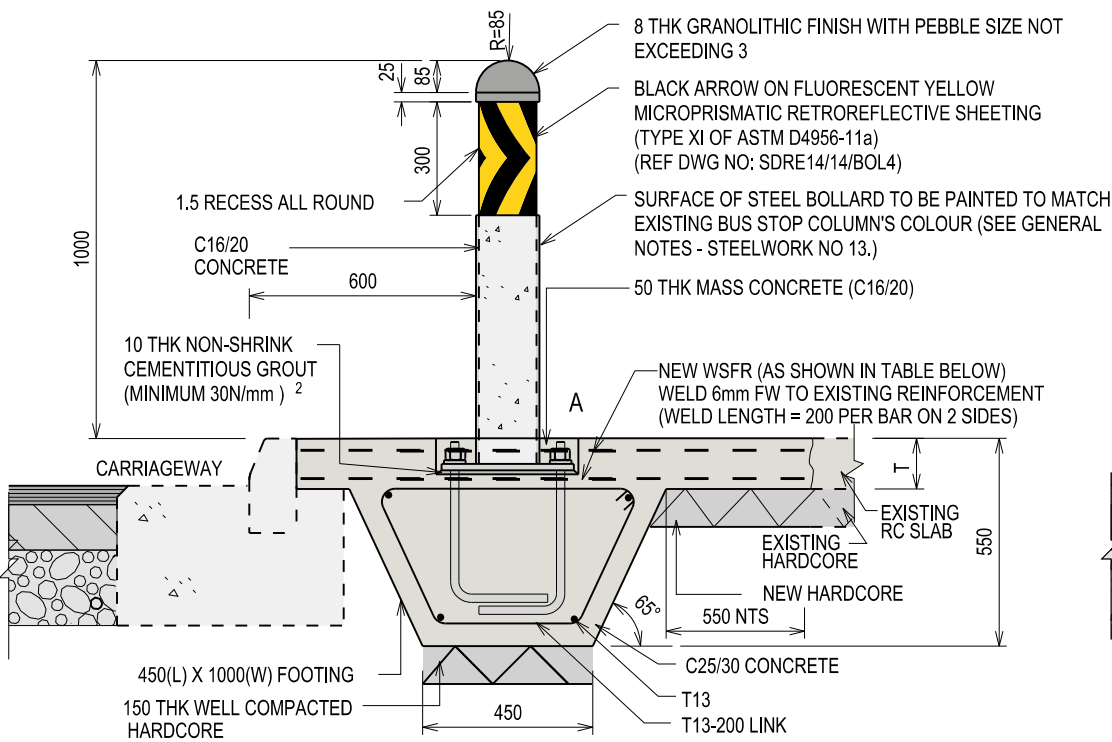
2. All steel reinforcement bars shall conform to BS 4449:1997 and SS 560.

STANDARD DETAIL

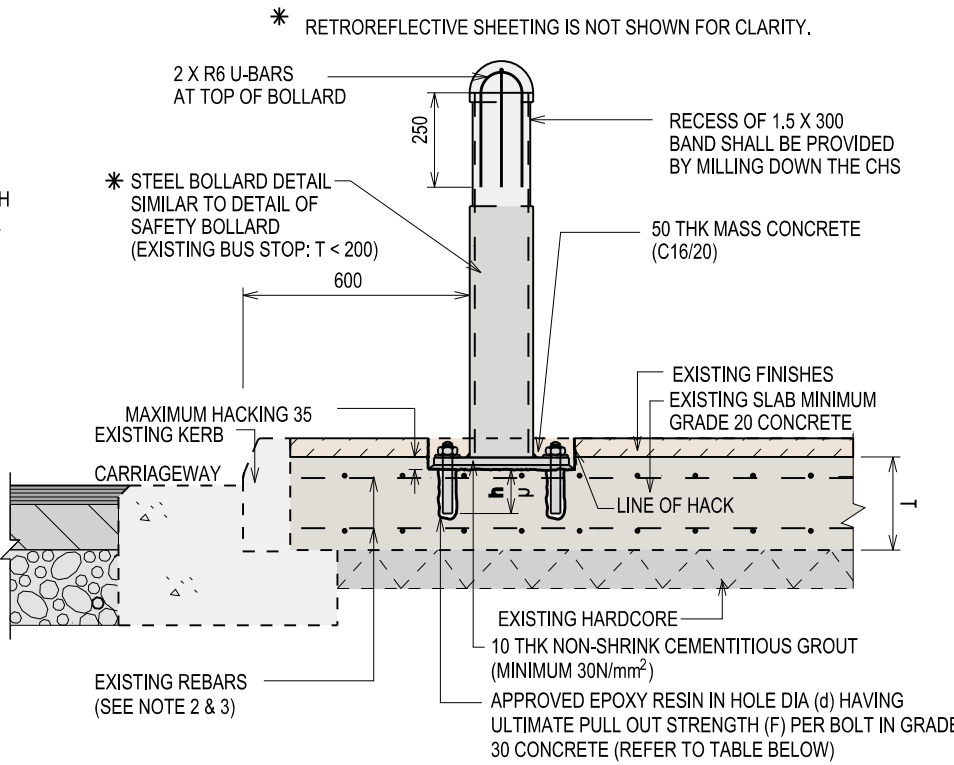
SPRING LOADED POST &
FLEXIBLE POST



DRAWING NO. LTA/SDRE14/14/BOL2		REV. A
DATE OF ISSUE 1st APR 2014	SCALE AS SHOWN	SHEET NO. 1 OF 1



CONNECTION DETAIL OF SAFETY BOLLARD
EXISTING BUS-STOP : 150mm ≤ T < 200mm
SCALE 1:20



CONNECTION DETAIL OF SAFETY BOLLARD
EXISTING BUS-STOP : T ≥ 200mm
DIMENSIONS h, d & F REFER TO TABLE IN DWG NO: SDR14/14/BOL4
SCALE 1:20

GENERAL NOTES:-
STEEL WORK

1. Fabrication of all structural steel work shall be carried out In accordance with SS EN 1993, BS EN 1090, SS EN 1994. All steel work shall be fabricated from new sections and in such manner that they are not bent, twisted or damaged.
2. Fabrication of all structural steel work including welding, trimming & painting shall be approved by S.O.
3. The material for all structural steel work shall comply with Parts 1 to 3 of BS EN 10025:2004, BS 7668:2004, BS EN 10029:2010, Parts 1 to 3 of BS EN 10210-1:2006
4. All steel section shall comply to CODE BS EN ISO 1461:1999 for hot-dipped galvanised and two coats of compatible paint shall be applied to all damaged surfaces.
5. All cut edges shall be dressed to a neat finish and shall be free from distortions.
6. All welding shall be in accordance with BS EN 1011-1:2009, BS EN 1011-2:2001 & electrodes to BS EN ISO 2560:2005.
7. No site welding shall be allowed unless approved by the S.O.
8. All steel bollards and base plates to be of GRADE S275 steel.
9. The general dimensions & tolerances of bolts shall comply with BS 4933:2010.
10. All bolts strength grade used are as specified in BS 4190:2014 GRADE 8.8 unless otherwise stated.
11. All bolts shall be fitted with washers and locking nuts complying with BS 4320:1968. Nuts shall be at least the strength grade appropriate to the grade of bolt used.
12. All stainless steel bolts shall comply to Part 1 to 2 of BS EN ISO 3506:2009 unless otherwise stated.
13. Prepare and apply one coat of approved two pack etching primer, one approved two pack epoxy undercoat (dry film thickness 150 microns) and two coats of approved two pack polyurethane paint (dry film thickness of 50 microns) to all galvanised metal surface and galvanised steel pipes to the satisfaction of the S.O.
14. All the welds between bollard and base plate shall be tested. 50% of the welds shall be tested by ultrasonic or radiographic tests and the other 50% of the welds shall be tested by magnetic particle flaw detection concrete.

CONCRETE

15. Minimum concrete cover to nearest reinforcement shall be 50mm for the foundation.
16. Concrete to be used shall have minimum concrete strength as specified in drawings.
17. All existing concrete surface shall be primed with bonding agent prior to casting of new concrete.

REINFORCEMENT

18. All steel reinforcement bars shall conform to BS 4449 and SS 560.
19. Reinforcement mill certificates must be submitted for acceptance prior to concreting work.

RETROREFLECTIVE SHEETING

20. Butt joint shall be used at the end of the retroreflective sheeting.
21. 10mm wide approved acrylic adhesive shall be applied along the butt joint.

NOTES FOR EXISTING BUS STOP

EXISTING SLAB THICKNESS	150mm ≤ T < 200mm	T ≥ 200mm
(1) Minimum concrete class	Concrete class at least C25/30 For existing slab, T< 150mm or concrete class C25/30 contractor shall proposed slab strengthening method to engineer's approval	Concrete class at least C25/30 For existing concrete class C25/30, Contractor shall proposed slab strengthening method to engineer's approval
(2) Minimum existing reinforcement area	Each layer of existing top & bottom reinforcement (fy =460N/mm ²) shall not be less than the areas specified below.	
	Posted speed 60 km/h	(a) 550mm ² /m (traffic direction) (b) 200mm ² /m (distribution bar)
	Posted speed 70 km/h	(a) 900mm ² /m (traffic direction) (b) 200mm ² /m (distribution bar)
(3) If item (2) is not satisfied	New reinforcement shall be added using either of the 2-method below: The added reinforcement shall cover an area of 1500mm x 1500mm. Main bars shall be placed parallel to traffic direction and nearer to the top surface & soffit of the slab for top & bottom reinforcement respectively.	
	Posted speed 60 km/h	(i) 2 layers of top & bottom reinforcement of WSFR A13 (ii) 1 top layer T13-100 (traffic direction), T10-200 (distribution bar)
	Posted speed 70 km/h	(i) 2 layers of top & bottom reinforcement of WSFR B13 (ii) 1 top layer T20-125 (traffic direction), T10-200 (distribution bar)

THE DIMENSION OF THE COMPONENTS IN THE DRAWINGS SHALL BE TAKEN AS THE MINIMUM SIZE REQUIRED.
THE SPECIFIED STRUCTURAL STRENGTH OF THE COMPONENTS SHALL BE CHECKED BY PROFESSIONAL ENGINEER

NOTES:

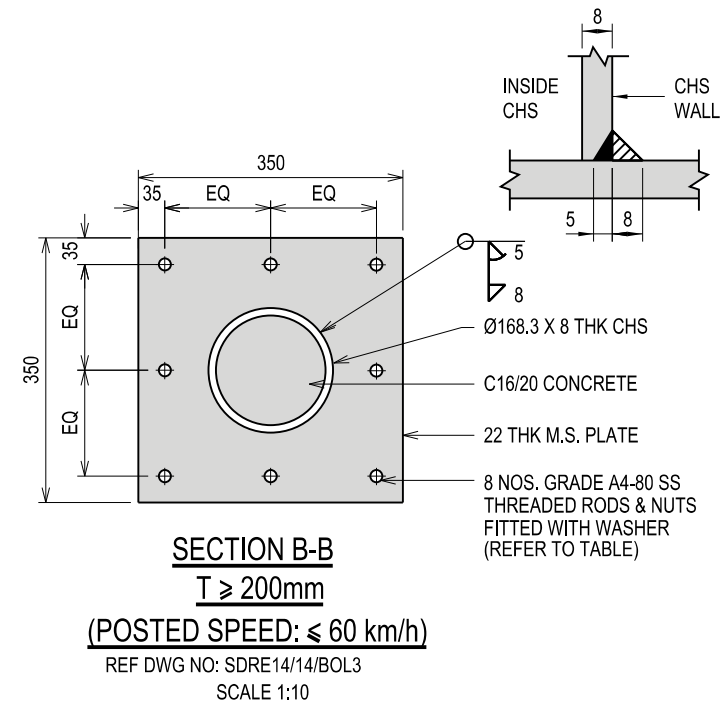
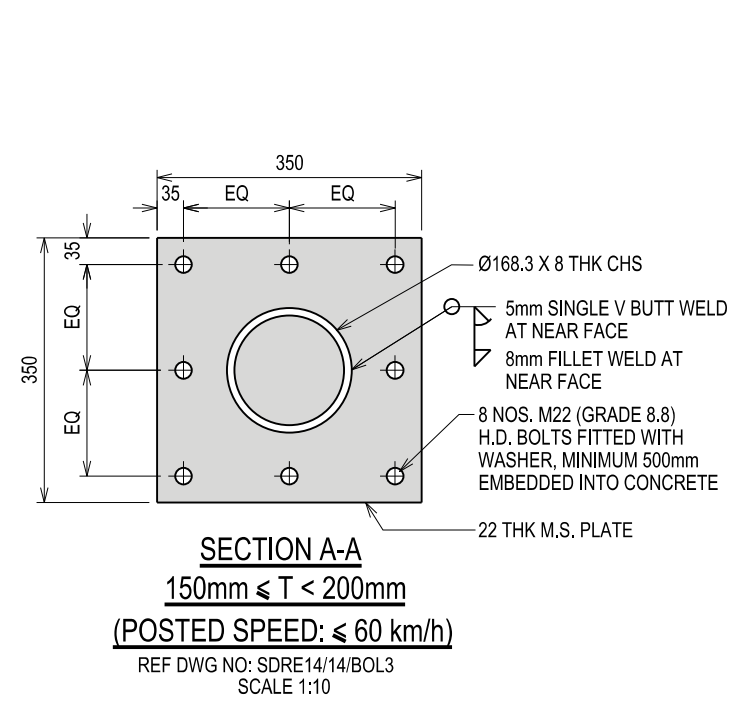
1. The structural details shown above are for bollard installed at bus-stop along road with posted speed limit of 60 km/h and 70 km/h.
2. Safety bollard installed within 2m from the perimeter of the new / upgraded bus-stop shall be welded to the reinforcement steel of the bus shelter platform to ensure equi-potential bonding is achieved and certified by licensed electrical worker.
3. Spacing of safety bollard to be at 3m c/c.

STANDARD DETAIL

SAFETY BOLLARD AT BUS STOP - POSTED SPEED :
60 km/h & 70 km/h
(SHEET 1 OF 2)

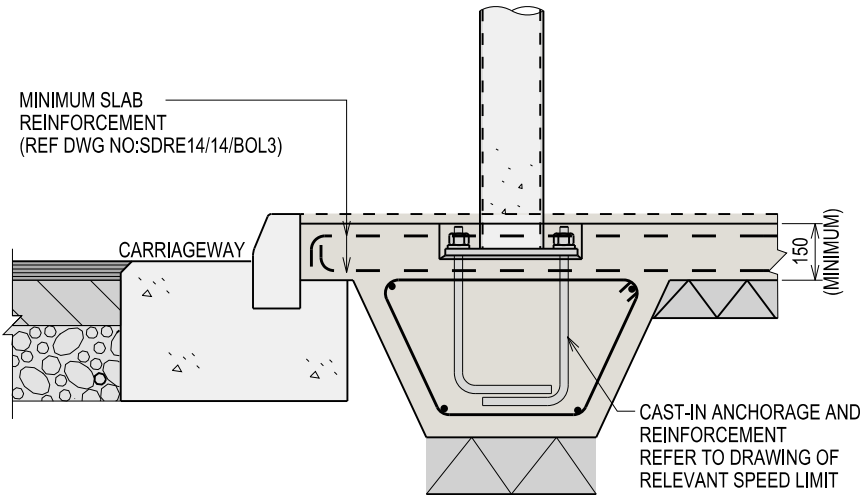


DRAWING NO.		REV.
LTA/SDRE14/14/BOL3		A
DATE OF ISSUE	SCALE	SHEET NO.
1st APR 2014	AS SHOWN	1 OF 2

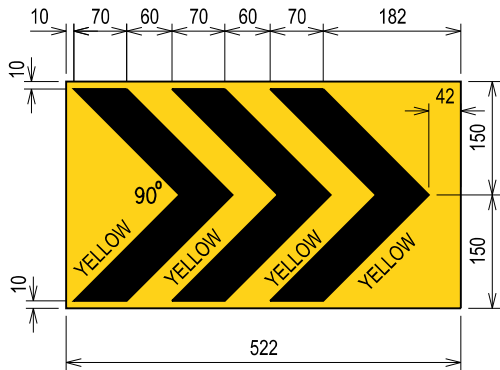
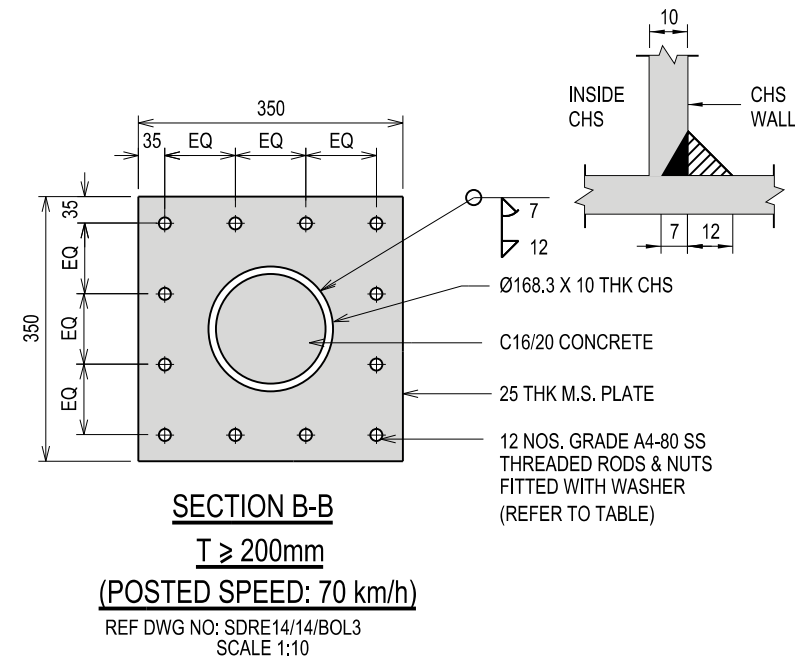
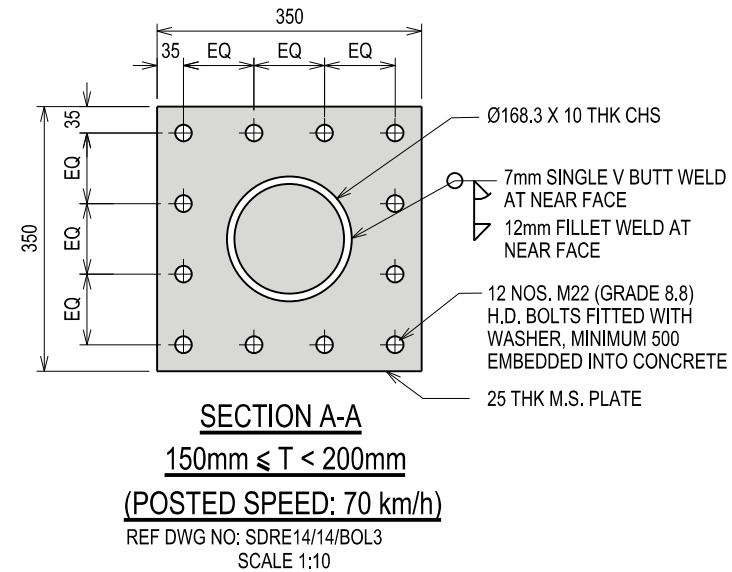


TABLE

POSTED SPEED	DIAMETER OF THREADED ROD (mm)	DIAMETER OF HOLE [d] (mm)	ULTIMATE PULL OUT STRENGTH [F] (kN)	ANCHOR DEPTH OF BOLT [h] (mm)
60 km/h	16	18	53	125
70 km/h	16	18	50	125

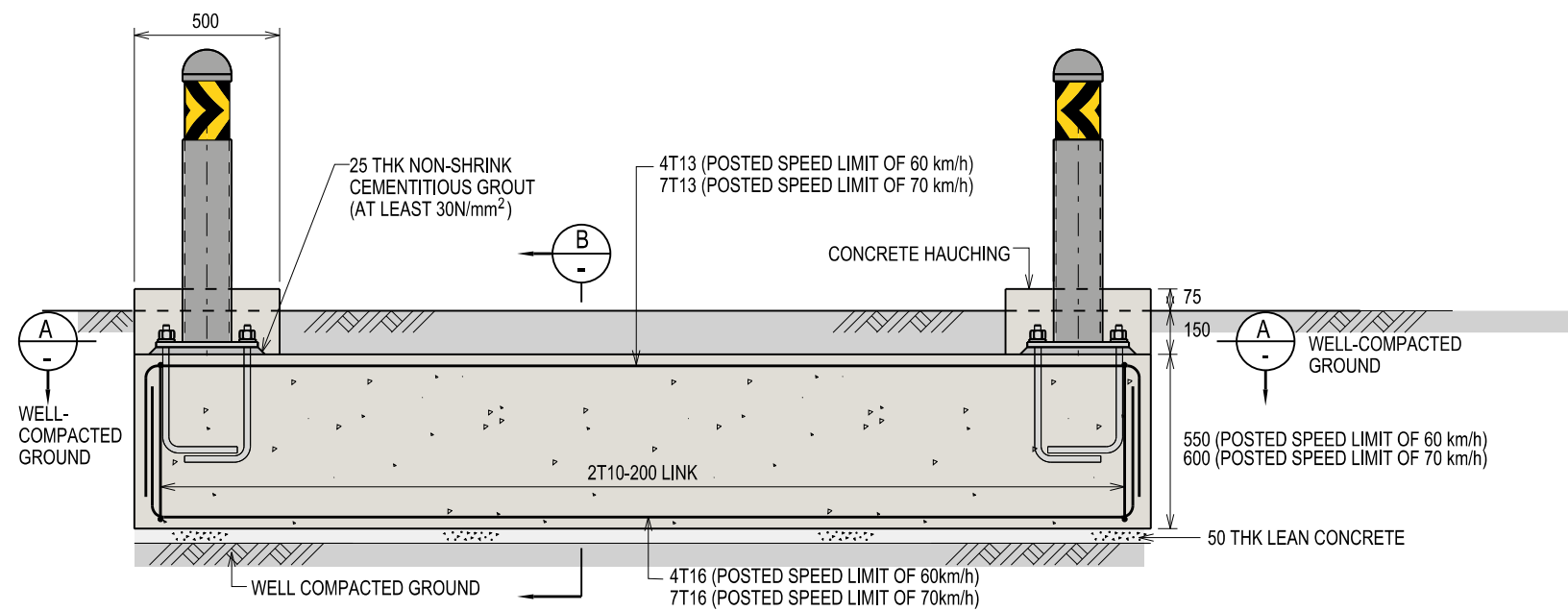


- NOTES:
- 1) FOR BOLLARD, FOOTING & PLATE DETAILS, REFER TO CONNECTION DETAIL OF SAFETY BOLLARD (EXISTING BUS STOP: $T < 200\text{mm}$)
 - 2) REFER TO NEW BUS STOP DRAWING FOR SLAB REINFORCEMENT DETAILS.

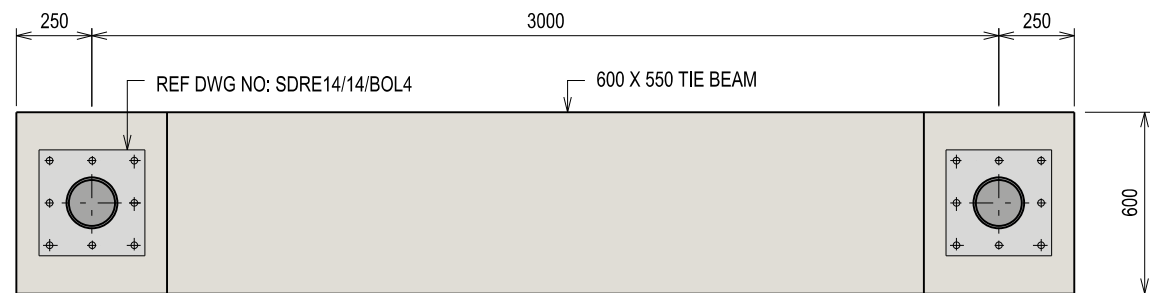


- NOTES:
1. For notes, refer to LTA/SDRE14/14/BOL3.

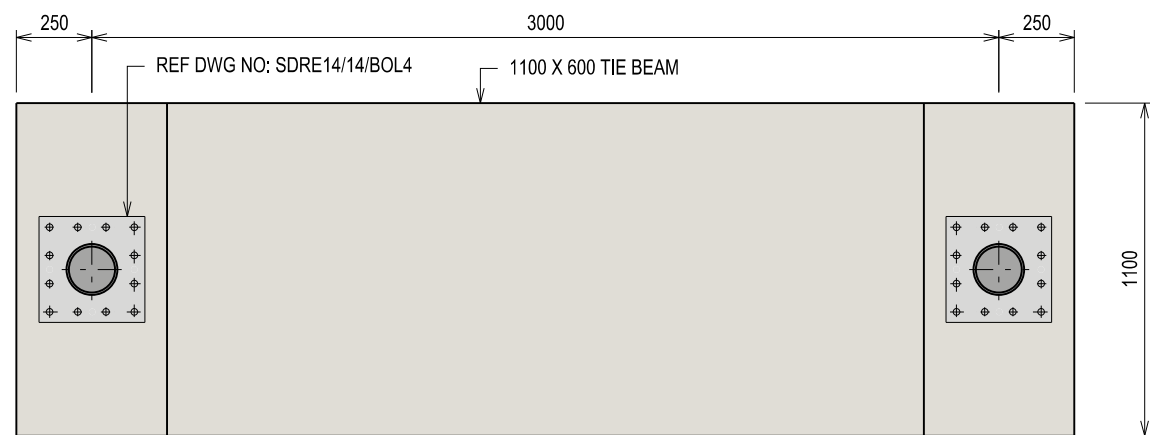
STANDARD DETAIL		Land Transport Authority	
SAFETY BOLLARD AT BUS STOP - POSTED SPEED : 60 km/h & 70 km/h (SHEET 2 OF 2)		DRAWING NO.	REV.
		LTA/SDRE14/14/BOL4	A
		DATE OF ISSUE	SHEET NO.
		1st APR 2014	2 OF 2
		SCALE	
		AS SHOWN	



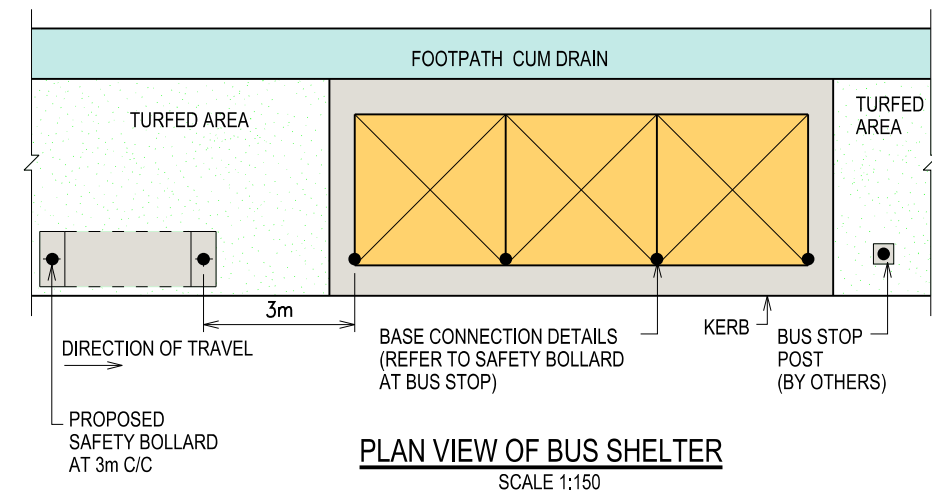
DETAIL OF BOLLARD AT TURF AREA
SCALE 1:25



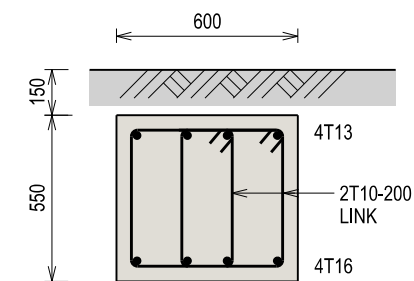
SECTION A-A
(POSTED SPEED LIMIT OF 60 km/h)
SCALE 1:25



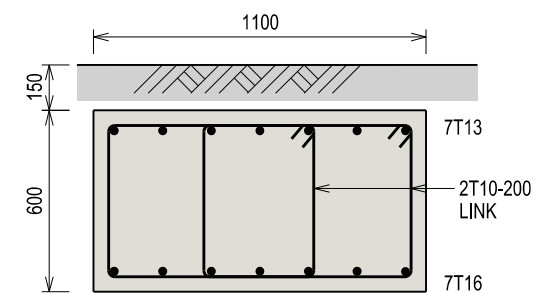
SECTION A-A
(POSTED SPEED LIMIT OF 70 km/h)
SCALE 1:25



PLAN VIEW OF BUS SHELTER
SCALE 1:150



SECTION B-B
(POSTED SPEED LIMIT OF 60 km/h)
SCALE 1:25



SECTION B-B
(POSTED SPEED LIMIT OF 70 km/h)
SCALE 1:25

NOTES:

1. For notes and details of bollard, refer to LTA/SDRE14/14/BOL3.
2. Tie beam shall be tied to at least 2 bollards.
3. Concrete used shall be of C28/35.
4. Ground surrounding the tie beam shall be properly compacted and any gap between the beam and the ground shall be grouted to Engineer's details or S.O.'s approval.

STANDARD DETAIL

SAFETY BOLLARD AT
TURFED AREA
- POSTED SPEED :
60 km/h & 70 km/h

Land Transport Authority

DRAWING NO.		REV.
LTA/SDRE14/14/BOL5		A
DATE OF ISSUE	SCALE	SHEET NO.
1st APR 2014	AS SHOWN	1 OF 1